Literature Review: Deep Learning-Based Fire and Smoke Detection (2026 Studies)

Ref	Paper Title	Author(s)/Year	Journal Name	Month & Year of Publication	Methodology & Technology	Advantages	Limitations
1	Deep Learning-Based Fire Detector Robust to Smoke–Fog Ambiguity in Outdoor Scenes	Suh, S. (2026)			YOLOv11 baseline extended with an added P6 image-pyramid layer and a novel lightweight C5Go module (ghost convolution); custom 3-class dataset (fire, smoke, fog, 3,100 images) built from FASDD and Roboflow sources.	Explicitly models fog as a hard-negative class, cutting smoke/fog false positives; +12.86% smoke mAP@50-95 over YOLOv11; modest compute overhead (+5.6% GFLOPs); mAP@50 of 0.906.	Custom dataset limited to 3,100 images; performance drops on external FASDD-CV dataset (domain shift); single-author study with limited real-world field validation.
2	Fire Detection Using Deep Learning	Kamble, D. & Khemnar, K. C. (2026)			CNN-based image/video classification pipeline using OpenCV frame processing from webcam/IP camera feeds; analyzes flame color, texture, brightness, and motion to classify Fire/No-Fire.	Simple, low-cost architecture; works with standard CCTV/webcam hardware; reduces reliance on delayed smoke/heat sensors; suited for continuous real-time monitoring.	Binary fire/no-fire classification only (no smoke or severity localization); no reported quantitative benchmarking against SOTA detectors; vulnerable to lighting/background variation.
3	An Optimized Deep Learning Framework for IoT-Based Smoke Detection with Enhanced Performance and Computational Efficiency	Nazish, M., Manan, A. & Banday, M. T. (2026)	Journal of Electrical Systems and Information Technology	April 2026	Optimized deep-learning framework integrated with IoT smoke sensor streams; designed for enhanced computational efficiency, with proposed model compression/pruning/quantization for edge deployment.	Improved computational efficiency suited to embedded/edge hardware; addresses sensor drift and reliability; framework designed for smart-infrastructure integration.	Relies on single-modality IoT sensor data (no image fusion yet proposed); real-world multi-site streaming validation left as future work; adaptive/self-learning mechanisms not yet implemented.
4	Intelligent IoT-Based Fire and Gas Leakage Detection System: Integrating Sensor Fusion, Edge AI, and Visual Verification	Altamash, M., Rayyan, M., Khan, S. A. & Parveen, H. (2026)	International Journal of Engineering Research & Technology (IJERT)	April 2026	Multi-sensor fusion (smoke, gas, temperature) combined with Edge AI processing and camera-based visual verification to reduce false alarms triggered by fixed sensor thresholds.	Sensor fusion plus visual confirmation lowers false alarms from environmental fluctuations; edge processing enables local, low-latency decisions; covers both fire and gas-leak hazards.	Increased system/hardware complexity and cost from multiple sensor types; edge AI accuracy depends on camera placement and lighting; limited large-scale deployment results reported.

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5	Real-Time Fire and Smoke Detection Using Vision Transformers and Spatiotemporal Learning	Lilhore, U. K. et al. (2026)			Hybrid vision transformer (ViT) architecture with self-attention fused with spatiotemporal learning to model visual patterns (color gradients, texture, intensity) across video frames.	Captures long-range spatial and temporal dependencies better than CNN-only models; strong accuracy on real-time video streams; robust to complex backgrounds.	Transformer architectures are computationally heavier than lightweight CNN/YOLO models, challenging low-power edge deployment; requires larger training data for stable attention learning.
6	An Intelligent Approach for Early Smoke/Fire Detection Using Vision Sensors in Smart Cities	Abozeid, A. & Alanazi, R. (2026)			ViT-based scene understanding paired with fast object detection; self-attention mechanism captures color gradient, texture, and intensity cues from smart-city vision sensor networks.	Strong reported accuracy and speed; well suited to urban safety applications needing rapid visual analysis; performs reliably under variable lighting and changing fire/smoke appearance.	Focused on urban/smart-city contexts, limiting generalization to forest or industrial settings; computational cost of transformer backbone may hinder deployment on constrained smart-city edge nodes.
7	Research on Fire Smoke Detection Algorithm Based on YOLOv11-FSNet	Chen, X., Zhang, B., Zhu, Y. et al. (2026)			Lightweight YOLOv11-FSNet with a C3k2-RVB-EMA backbone/neck module (depthwise separable convolution + dynamic attention), a Feature-Focused Diffusion Pyramid Network (FSNet) neck, and a Mixed Local Channel Attention (MLCA) mechanism.	Improves mAP@0.5, precision, and recall (+1.6%, +2.5%, +1.7%) over baseline; enhanced sensitivity to ambiguous, diffuse smoke targets; reduced computational cost via lightweight modules.	Gains are incremental over baseline YOLOv11; performance in complex multi-scale outdoor scenes with occlusion not fully detailed; limited cross-dataset generalization evidence.
8	Real-Time Smart Fire Detection and Prevention System Using IoT Sensors and ML Algorithms	Swetha, P. B., Vennakandla, C. S. R., Sonti, S. R., Periasamy, P. S., Naresh, A. & Gali, S. (2026)			IoT sensor network (smoke, temperature, gas) combined with machine learning classification algorithms for real-time fire prediction and automated prevention/alerting.	Enables proactive prevention, not just detection; integrates multiple low-cost IoT sensors; real-time alerting suited to homes/small buildings.	ML classifiers reported are simpler than deep vision models, limiting accuracy in visually complex scenes; no visual/image-based confirmation layer to reduce sensor-driven false alarms.

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9	Fire-YOLO26: An NMS-Free Dual-Stream RGB–Thermal Network for Real-Time UAV Forest Fire Detection in Smoke-Challenging Conditions	Zhang, S., Chen, Z., Xiyang, J. et al. (2026)			NMS-free dual-stream RGB–thermal infrared (TIR) network built on YOLO26; Thermal-guided Feature Alignment Module (T-FAM) recalibrates RGB features using heat-driven spatial masks; Dual-modal Small Target Aware Labeling (Dual-STAL) for early small-fire learning.	Removes NMS-induced latency for faster UAV inference; RGB-thermal fusion suppresses smoke occlusion and fire-like background noise; strong small-fire and early-detection capability.	Requires dual RGB-thermal camera hardware, raising UAV payload/cost; performance under extreme thermal distractor conditions (e.g., hot machinery) not fully characterized.
10	A Fire Detection Method Based on a Mind-Linked Continuous-Coupled Neural Network	Liu, K., Wang, J., Yang, W. et al. (2026)			Brain-inspired mind-linked continuous-coupled neural network (ML-CCNN) built on continuous-coupled neural network (CCNN) theory; parameter-adaptation via global threshold modulation; SMOTE used for multi-sensor data-imbalance correction.	Biologically inspired design improves robustness to environmental noise and sensor heterogeneity; reduces false alarms/missed detections versus conventional ML classifiers; addresses class imbalance directly.	Relies purely on multi-sensor (non-visual) data, lacking image-based confirmation; novel CCNN-based architecture is less interpretable/benchmarked than mainstream CNN/YOLO approaches.

Research Gap

Synthesizing the limitations reported across the ten reviewed studies (2026), the following gaps emerge in current deep learning-based fire and smoke detection research:

- **Limited multimodal fusion.** Most studies rely on a single modality — either pure vision (YOLO/ViT-based, Refs 1, 5–9) or pure sensor data (IoT/CCNN-based, Refs 3, 4, 8, 10) — with few systems combining visual confirmation and sensor fusion together (only Ref 4 does both meaningfully). This leaves most approaches vulnerable to their single modality's blind spots (e.g., visual occlusion by smoke, or sensor drift/false triggers).
- **Insufficient real-world, large-scale validation.** Several works (Refs 1, 3, 4, 7, 8, 9) explicitly flag that real-world, multi-site, or large-scale field deployment results are limited or left as future work; most report only lab-scale or benchmark-dataset performance.
- **Weak cross-dataset / domain generalization.** Custom or narrow datasets (e.g., Ref 1's 3,100-image set) show performance drops under domain shift when tested on external datasets (e.g., FASDD-CV), and several transformer-based models (Refs 5, 6) require larger training sets for stable generalization.
- **Accuracy vs. edge-efficiency trade-off unresolved.** Transformer-based architectures (Refs 5, 6) achieve strong accuracy but are computationally heavy for low-power edge/UAV deployment, while lightweight IoT/CNN approaches (Refs 2, 3, 8) sacrifice detection granularity (e.g., binary fire/no-fire only, no severity or localization).
- **Lack of adaptive / self-learning mechanisms.** Sensor-drift correction, adaptive thresholds, and continual/self-learning capability are explicitly noted as unimplemented (Ref 3) or absent (Ref 8), leaving most systems static once deployed.
- **Rising hardware cost and complexity.** Multi-sensor fusion and dual-modality systems (Refs 4, 9) improve reliability but increase hardware cost, payload, and deployment complexity — a barrier to large-scale rollout, especially for UAV-based systems.
- **Limited interpretability of novel architectures.** Biologically-inspired or highly customized architectures (Ref 10's ML-CCNN) are less interpretable and less benchmarked against mainstream CNN/YOLO baselines, making comparative evaluation difficult.
- **No unified benchmarking standard.** Across the reviewed studies, evaluation uses different datasets, metrics, and hardware baselines, making direct cross-paper comparison of accuracy/efficiency claims difficult and highlighting the need for a standardized fire/smoke detection benchmark.

SDG Mapping

The reviewed literature aligns with the following United Nations Sustainable Development Goals (SDGs):

SDG	Relevant Studies	Mapping Rationale
SDG 3 – Good Health and Well-Being	Refs 1–10 (all)	Faster, more accurate fire/smoke detection directly reduces injury, death, and health impacts from fire and smoke inhalation by enabling earlier warning and response.
SDG 9 – Industry, Innovation and Infrastructure	Refs 1, 3–5, 7, 9, 10	Advances in YOLO variants, vision transformers, edge AI, and IoT-based sensor frameworks represent innovation in resilient infrastructure and industrial safety technology.
SDG 11 – Sustainable Cities and Communities	Refs 2, 3, 4, 6, 8	IoT-based smoke/gas detection, smart-city vision sensor networks, and home/building fire prevention systems directly support safer, more resilient urban and community infrastructure.
SDG 13 – Climate Action	Refs 1, 5, 7, 9	Improved outdoor and forest-adjacent fire/smoke detection (including fog/smoke ambiguity handling) supports mitigation of fire-driven greenhouse gas emissions and climate-related disaster risk.
SDG 15 – Life on Land	Refs 1, 9	UAV- and outdoor-scene-based forest fire detection (e.g., Fire-YOLO26's UAV RGB-thermal system) directly supports early wildfire detection, protecting forest ecosystems and terrestrial biodiversity.

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